Second-Order IVPs and First-Order Systems

1.] The undamped mass-on-a-spring system is modeled by the second-order differential equation

$$m\frac{d^2u}{dt^2} + ku = 0, \qquad u(0) = u_0, \quad u'(0) = v_0$$

where $m > 0$ is the mass, $k > 0$ is the spring constant, and $u(t)$ is the displacement from equilibrium. Write this system as a first-order system of ODEs, then solve it in Matlab for various values of m and k. Make a second plot that plots v versus u. This is called the *Phase Plane*, which can be plotted on the vector field.

2.] The damped mass-on-a-spring system is modeled by the second-order differential equation

$$m\frac{d^2u}{dt^2} + b\frac{du}{dt} + ku = 0, \qquad u(0) = u_0, \quad u'(0) = v_0$$

where $m > 0$ is the mass, $b \geq 0$ is the damping coefficient, $k > 0$ is the spring constant, and $u(t)$ is the displacement from equilibrium. This equation describes oscillatory motion where the damping force opposes the velocity, causing the oscillations to gradually decrease in amplitude over time. Write this system as a first-order system of ODEs, then solve it in Matlab for various values of m, b, and k. Plot the time-series, vector field, and Phase Plane.

3.] The Lotka-Volterra equations describe a *predator-prey* relationship:

$$\begin{aligned}
\frac{du}{dt} &= \alpha u - \beta uv \qquad & u(0) = u_0 \\
\frac{dv}{dt} &= -\gamma v + \delta uv \qquad & v(0) = v_0
\end{aligned}$$

Choose a set of parameters for this model and solve the system in Matlab. Plot the time-series, vector field, and Phase Plane.

4.] Below is a two-dimensional first-order system of equations that models *competing species*:

$$\begin{aligned}
\frac{du}{dt} &= k_1 u \left(1 - \frac{u}{m_1}\right) - \delta_1 uv \qquad & u(0) = u_0 \\
\frac{dv}{dt} &= k_2 v \left(1 - \frac{v}{m_2}\right) - \delta_2 uv \qquad & v(0) = v_0
\end{aligned}$$

Choose a set of parameters for this model and solve the system in Matlab. Plot the solutions against time. Plot the time-series, vector field, and Phase Plane.